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OUTLINER HAIR TRIMMER WITH ENHANCED WATER RESISTANT SEALS

Abstract

A hair trimmer includes a trimmer housing having a drive end, a drive shaft projecting from the drive end, a chassis having one end associated with the drive shaft and an opposite end having at least one projecting charging pin, a radial flange at each of the ends, an endcap at each of the ends, each endcap having an axially extending annular skirt, the annular skirt dimensioned to enclose the corresponding chassis end, and a tubular sleeve of heat deformable plastic film dimensioned for slidably surrounding the chassis and extending axially at least to the flanges, such that upon application of heat, the sleeve deforms around the chassis and the endcaps and forms a watertight enclosure.

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Background/Summary

BACKGROUND

[0001] The present invention relates generally to electric hair trimmers, and more specifically to such devices used for achieving precision and detailed trimming styles on the customer's scalp and/or face, or trimming nose and ear hair.

[0002] Modern hair grooming is more popular and often features design elements cut into the customer's hair down to the skin. These design elements take a variety of forms including but not limited to well-defined edges for facial hair, geometric designs, freeform patterns, images of animals, text and the like. Also, modern grooming includes maintenance for areas like nose and ear hair. Precision grooming is often challenging to home hair groomers using conventional electric hair trimmers, where it is often difficult to view the cutting blade and area, since traditional hair clippers have housings that extend to the bladeset cutting teeth. As such, the housing often obscures the view of the cutting area to achieve precision.

[0003] Hair clipper and trimmer manufacturers have addressed this need by providing specialized outliner trimmers (in the present application, hair clippers and hair trimmers are collectively referred to as hair trimmers) which have an exposed bladeset positioned axially from an end of the housing for enhancing the hair groomer or stylist's view of the cutting area. Designers of such outliner trimmers are forced to balance the often competing demands of enhanced visibility of the cutting area and the requirement for stabilized mechanical and electrical connection of the bladeset to the clipper drive system.

[0004] Still another challenge to hair grooming, besides creating precision modern hairstyles with such design shapes, is that the customer often needs other portions of his hair trimmed in a more conventional manner. Thus, the groomer or stylist creating a complete grooming experience for the customer should be equipped with a variety of powered grooming appliances for different grooming tasks. Since the modern trend is away from corded appliances to rechargeable cordless models, the stylist often is faced with managing a crowded workspace filled with various appliances and their respective charging stands.

[0005] For customers who prefer to create and/or maintain their stylized hair at home, there is a resistance to maintaining a variety of hair cutting appliances. Also, there is a growing demand for such devices that are rinseable for cleaning or usable in the shower or bathtub. Designers of hair trimmers are challenged with addressing this new demand.

[0006] Accordingly, there is a need for an improved outliner trimmer which addresses the above-listed hair trimmer design considerations.

SUMMARY

[0007] The above-listed need is met or exceeded by the present outliner hair trimmer with interchangeable workheads. Featuring an exposed, axially-extending drive shaft that projects from the trimmer housing, the present trimmer also provides a workhead attachment system that allows the user to interchangeably connect one of a plurality of workheads with the drive shaft without the use of tools. In the present application, "without the use of tools" means that the structure for connecting the cutting workhead to the trimmer housing is constructed such that it is free from any structure that is designed to accept a tool to manipulate the same, i.e. it is constructed for hand manipulation. Each of the workheads, preferably including a nose trimmer, a dry shaver and a hair trimmer, features an exposed, highly visible bladeset which enhances the user's ability to view the area being cut free of obstruction from the surrounding housing.

[0008] Once the user desires to change workheads, in one embodiment, unscrewing a threaded thumbscrew allows easy release of a support bracket from the trimmer housing and disconnection of the mounted workhead, and exchange of another workhead, which is engageable with the drive shaft and is releasably secured to the trimmer housing using the thumbscrew. In an embodiment, a compressible member or bushing is interposed between a gripping portion of the thumbscrew and a contact surface of the workhead bracket to provide user enhanced tactile feedback of tightness.

[0009] In another embodiment, instead of the above-described thumbscrew, the support bracket is provided in two components, one, a base portion secured to the workhead and another, a spring clip portion secured to the trimmer housing and having a complementary engagement structure engaging the base portion. Preferably, complementary formations on the two components enhance the connection of the workhead to the housing. In one embodiment, the spring clip portion includes two spring tabs exerting a biasing force on the base portion, and at least one lug matingly engaging an opening in the base portion. As such, the workhead is held to the trimmer through engagement of the drive shaft, as well as through multiple attachment points associated with the support bracket.

[0010] Still another feature of the present trimmer is that the inner drive system, motor and battery are waterproofed so that the trimmer is usable in the shower or bathtub. The waterproofing is obtained by inserting an inner trimmer chassis into a tubular sleeve of plastic film. In one embodiment, the plastic film is shrink wrap film. Radially extending disk-like flanges are disposed at each end of the chassis. As heat is applied, the shrink wrap film collapses around the trimmer components, sealing them from moisture, and the flanges become anchor points for ends of the tubular sleeve. Charging pins project from one of the flanges.

[0011] At the opposite end of the chassis, a motor shaft seal prevents the entry of water into the motor. It is also preferred that a segment of resilient tubing is placed around a main on-off switch of the trimmer prior to the application of heat to the shrink wrap film. The tubing segment surrounds and protects the switch as the film collapses about the chassis, and creates a flat surface around the switch for the shrinking film. In another embodiment, the film is secured to the chassis by welding or the like at the enlarged flanges.

[0012] More specifically, a hair trimmer includes a trimmer housing having a drive end, a drive shaft projecting from the drive end, a chassis having one end associated with the drive shaft and an opposite end having at least one projecting charging pin, a radial flange at each of the ends, and a tubular sleeve of heat deformable plastic film dimensioned for slidably surrounding the chassis and extending axially at least to the flanges, such that upon application of heat, said sleeve deforms around the chassis and forms a watertight enclosure.

[0013] In an embodiment, the flanges form sealing attachment points for the tubular sleeve, wherein the sleeve is attached to the chassis by one of heat shrinking and welding to the flanges. In another embodiment, a switch is secured to the housing, and an annular protector surrounding the switch and forming a guide for, and provide a flat profile for the switch as the sleeve is heated and sealingly engages portions of the trimmer.

[0014] In an embodiment, the chassis is provided with a pair of endcaps, each placed, prior to installation of the tubular sleeve at a corresponding end of the chassis and including a skirt projecting axially along the chassis to form a bonding point for the tubular sleeve.

[0015] In an embodiment at the drive shaft end, the endcap is provided with an opening accommodating the drive shaft and a drive shaft seal. Preferably, at each endcap, the skirt extends axially at least one inch.

[0016] In another embodiment, a hair trimmer includes a trimmer housing having a drive end, a drive shaft projecting from the drive end, a chassis having one end associated with the drive shaft and an opposite end having at least one projecting charging pin, a radial flange at each of the ends, an endcap at each of the ends, each endcap having an end wall from which axially extends an annular skirt, the annular skirt dimensioned to enclose the corresponding chassis end, and a tubular sleeve of heat deformable plastic film dimensioned for slidably surrounding the chassis and extending axially at least to the flanges, such that upon application of heat, the sleeve deforms around the chassis and the endcaps and forms a watertight enclosure.

[0017] In another embodiment, an encased drive module includes a chassis having one end associated with a drive shaft and an opposite end having at least one projecting charging pin, a radial flange at each of the ends, an endcap of resilient material enclosing each end of the chassis, a

tubular sleeve of heat deformable plastic shrink wrap film dimensioned for slidingly surrounding the chassis and the endcaps and extending axially at least to the flanges, such that upon application of heat, the sleeve deforms and shrinks around the chassis and said endcaps and forms a watertight enclosure.

[0018] In an embodiment, the chassis includes a plurality of integrally formed mounting formations for retaining components mounted to the chassis, and wherein the tubular sleeve attaches to and seals around the flanges and the components mounted to the chassis.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] FIG. 1 is a top perspective view of the present hair trimmer equipped with a nose trimmer workhead;

[0020] FIG. 2 is a fragmentary exploded perspective view of the hair trimmer of FIG. 1;

[0021] FIG. 3 is a fragmentary vertical cross-section of the hair trimmer of FIG. 1;

[0022] FIG. 4 is a side elevation of the nose trimmer workhead and support bracket;

[0023] FIG. 5 is a fragmentary top perspective view of the present hair trimmer equipped with a dry shaver workhead or foil that is usable with water;

[0024] FIG. 6 is an exploded perspective view of the hair trimmer of FIG. 5;

[0025] FIG. 7 is a vertical cross-section of the hair trimmer of FIG. 5;

[0026] FIG. 8 is a side elevation of the hair trimmer of FIG. 5;

[0027] FIG. 9 is a fragmentary top perspective view of the present hair trimmer equipped with a hair trimmer workhead;

[0028] FIG. 10 is an exploded perspective view of the hair trimmer of FIG. 9;

[0029] FIG. 11 is a vertical cross-section of the hair trimmer of FIG. 9;

[0030] FIG. 12 is a side elevation of the hair trimmer of FIG. 9;

[0031] FIG. 13 is a fragmentary top perspective view of the present hair trimmer equipped with a nose trimmer workhead and a spring-clip support bracket;

[0032] FIG. 14 is a fragmentary exploded perspective view of the hair trimmer of FIG. 13;

[0033] FIG. 15 is a vertical cross-section of the hair trimmer of FIG. 13;

[0034] FIG. 16 is a side elevation of the hair trimmer of FIG. 13;

[0035] FIG. 17 is a fragmentary top perspective view of the present hair trimmer with a dry shaver workhead and a spring-clip support bracket;

[0036] FIG. 18 is a fragmentary exploded view of the hair trimmer of FIG. 17;

[0037] FIG. 19 is a vertical cross-section of the hair trimmer of FIG. 17;

[0038] FIG. 20 is a side elevation of the hair trimmer of FIG. 17;

[0039] FIG. 21 is a fragmentary top perspective view of the present hair trimmer mounted using a spring-clip support bracket;

[0040] FIG. 22 is a fragmentary exploded view of the hair trimmer of FIG. 21;

[0041] FIG. 23 is a vertical cross-section of the hair trimmer of FIG. 21;

[0042] FIG. 24 is a side elevation of the hair trimmer of FIG. 21;

[0043] FIGS. 25-27 are sequential views of the removal of the workhead of FIG. 21 from the present hair trimmer;

[0044] FIG. 28 is a top perspective view of a chassis of the present hair trimmer surrounded by a tubular sleeve of heat deformable plastic;

[0045] FIG. 29 is a top perspective view of the chassis of FIG. 28 after the application of heat;

[0046] FIG. 30 is an end top perspective view of the chassis of FIG. 29;

[0047] FIG. 31 is a rotated top view of the chassis of FIG. 29;

[0048] FIG. 32 is a fragmentary end view of the chassis of FIG. 29;

[0049] FIG. **33** is a fragmentary side elevation of an embodiment of the chassis of FIG. **29**;
[0050] FIG. **34** is a side elevation of the chassis of FIG. **28** prior to the application of the tubular sleeve;
[0051] FIG. **35** is an exploded perspective view of the present hair trimmer and chassis;
[0052] FIG. **36** is a top view of an alternate embodiment of the plastic sleeve used to waterproof a chassis of the present trimmer;
[0053] FIG. **37** is an exploded perspective view of an alternate embodiment of the hair trimmer of FIG. **1**;
[0054] FIG. **38** is a perspective view of the bushing seen in the hair trimmer of FIG. **37**;
[0055] FIG. **39** is a partially exploded top view of another embodiment of the present trimmer motor pod;
[0056] FIG. **40** is an assembled top of view of the motor pod embodiment of FIG. **39** shown assembled with a shrink wrap sleeve encompassing the motor pod; and
[0057] FIG. **41** is an exploded top perspective view of the motor pod embodiment of FIG. **39**.

DETAILED DESCRIPTION

[0058] Referring now to FIGS. **1-4**, an outliner hair trimmer is generally designated **10**, and includes a trimmer housing **12** having a drive end **14**. A feature of the present hair trimmer **10** is that it is configured for operationally engaging a plurality of cutting workheads. The first such cutting workhead is a nose trimmer, generally designated **16**.

[0059] The nose trimmer workhead **16** engages a drive shaft **18** that projects from the drive end **14** and is connected to an offset or eccentric cam **20** having a polygonal, preferably hexagonal, drive formation **22**. In the nose trimmer workhead **16**, a cam follower **24** has a base **26** with an internal recess **28** that matingly engages the drive formation **22** for common rotation by the drive shaft **18**, being driven by a motor **30** (FIG. **28**) as is known in the art. A tubular, moving blade **32** having a plurality of teeth **34** is drivingly engaged by, and rotates with the cam follower **24** (FIG. **3**). A knurled collar **36** secures a tubular outer or stationary blade **38** having a plurality of stationary blade teeth **40** in relation to the cam follower **24** so that the moving blade **32** rotates internally of, and relative to the stationary blade **38** as is known in the art.

[0060] Hairs extending internally past the stationary blade teeth **40** are sheared by the action of the rotating moving blade teeth **34**. By extending axially from the drive end **14** of the housing **12**, the nose trimmer workhead **16** is relatively exposed and as such provides enhanced user visibility of the target cutting area.

[0061] To provide structural support to the nose trimmer workhead **16**, and also to provide a releasable connection with the trimmer housing **12**, a support bracket **42** is associated with the drive end **14** of the housing. The support bracket **42** is also connected to the cutting workhead **16**. More specifically, the present support bracket **42** has a first end **44** configured for engaging the nose trimmer workhead **16**, preferably having a loop **46** at least partially surrounding the stationary blade **38**. The loop **46** is held in place against the stationary blade **38** by the knurled collar **36**. Opposite the first end **44**, a second end **48** is configured for engaging the clipper housing **12**. In the preferred embodiment, the second support bracket end **48** has a clevis shape **50** defining a slot **52** (FIG. **2**) that slidingly engages a threaded female insert **53** and thumbscrew **54** with a knurled or otherwise grip-enhanced knob **56**. Also preferred, the support bracket **42** has a mid-portion **58** forming an angle for enhanced strength.

[0062] To remove the nose trimmer workhead **16** from the trimmer housing **12**, the user manually unscrews the thumbscrew **54** and axially pulls the workhead out of engagement with both the thumbscrew as well as the eccentric cam **20**. Installation of the nose trimmer workhead **16** is achieved by reversing the above process. Thus, the nose trimmer workhead **16** is removable from operational engagement with the trimmer housing **12** without the use of tools.

[0063] Referring now to FIGS. **5-8**, another cutting workhead that is readily disengageable from the trimmer housing **12** is generally referred to as a dry shaver or shaver workhead **60**. Although

referred to herein as a “dry” shaver, the present workhead **60** is usable with water. The shaver workhead **60** includes a workhead housing **62** that is configured with a relatively slim profile to enhance user visibility of the cutting area. As such, a sidewall **63** is relatively short vertically and is shorter than a relatively wider, flared end described below as **88**. Included in the workhead housing **62** is a pivoting rocker arm **64** having a cam follower socket **66** that engages the eccentric cam **20** through an opening **68** in the workhead housing. The opening **68** defines an operational channel for the pivoting action of the rocker arm **64**. Reciprocally pivoting relative to the workhead housing **62** about a pivot point **70**, opposite the cam follower socket **66**, a free end **72** of the rocker arm **64** engages moving shaver blades **74** to linearly reciprocate them relative to the fixed, perforated, stationary blade or foil **76**. A bias element **77** such as a spring is positioned between a rear end **73** of the rocker arm **64** opposite the free end **72** and an adjacent portion of the opening **68** (FIG. 7) and, through biasing action on the rocker arm, urges the moving blades **74** against the stationary blade **76**. As is well known in the art, the direction of linear reciprocation of the moving shaver blades **74** is transverse to a longitudinal axis of the trimmer housing **12**. Hairs which project into the perforated stationary blade **76** are clipped by the reciprocating moving shaver blades **74**.

[0064] In similar fashion to the nose trimmer workhead **16**, the dry shaver cutting workhead **60** is releasably operationally engageable with the trimmer housing **12** through the use of a support bracket **78**, which is similar to the support bracket **42** in that a second end **48** has a clevis shape **50** with a slot **52**. The clevis shape **50** releasably engages the thumbscrew **54** in the same manner as described above in relation to the nose trimmer workhead **16**. In addition to the fastening power of the thumbscrew **54**, the drive end **14** of the trimmer housing is provided with a polygonal clevis recess **80** that captures the second end **48** and prevents twisting of the bracket **42**, **78** relative to the housing.

[0065] Opposite the second end **48**, a first end **82** of the support bracket **78** is securely fastened to the workhead housing **62** using at least one, and preferably a pair of fasteners **84**. An upper edge **86** of the support bracket **78** is angled to engage a flared blade end **88** of the workhead housing **62**, and the mid portion **58** has a similar angle as is the case with the nose trimmer workhead **16**. The upper edge **86** also is removable from the workhead **60** so that any trapped hair can be removed from an interior of the workhead. While the present upper edge **86** is integral with the first end **82**, it is also contemplated that the upper edge is optionally provided as a separate door structure that is removable without removing the fasteners **84**.

[0066] Referring now to FIGS. 9-12, a third type of cutter workhead is generally designated a hair trimmer workhead **90**. Upon mounting to the drive end **14** of the trimmer housing **12** and engagement upon the drive shaft **18**, the hair trimmer workhead **90** provides an exposed bladeset **92** including a stationary blade **94** having stationary blade teeth **96**, and a laterally reciprocating moving blade **98** with moving blade teeth **100**. As such, the user or stylist can more clearly view the area being cut, for achieving a desired precision and detail cut according to current fashion.

[0067] As is the case with the nose trimmer workhead **16** and the dry shaver workhead **60**, the hair trimmer workhead **90** is connected to the trimmer housing **12** by a support bracket **102**. A second end **48** of the support bracket has the clevis shape **50** defining the slot **52** for releasable engagement with the thumbscrew **54** as described above. The second end **48** is received or captured within the clevis recess **80** in the trimmer housing **12**. Further, the support bracket **102** has the angled mid-portion **58** in similar fashion to the previously described support brackets **42**, **78**.

[0068] Referring now to FIG. 10, opposite the second end **48**, the support bracket **102** has a first end **104** configured for attachment to the hair trimmer workhead **90**. A connector **106** features bores **108** for receiving fasteners **110**, preferably screws or the like, which hold the stationary blade **94** and a blade guide **112** together as a unit. Also, the fasteners **110** hold the stationary blade **94**, the blade guide **112** and the connector **106** to the first end **104** via engagement in threaded openings **114**. A depending lug **116** on the connector also matingly engages a slot **118** in the first end **104** of the support bracket **102** for stabilizing the workhead **90** during operation.

[0069] As is well known in the art relating to hair trimmers, the hair trimmer workhead **90** includes a cam follower **120** with a socket **122** constructed and arranged for engaging the eccentric cam **20**. Prongs **124** on the cam follower **120** engage openings **126** in the blade guide **112**, and formations **128** on the blade guide engage a central opening **130** in the moving blade **98** to maintain the lateral reciprocating trajectory of the moving blade relative to the stationary blade **94**. A biasing element **132** such as a spring urges the cam follower **120** and the moving blade **98** against the stationary blade **94**. In performing detailed trimming with the hair trimmer workhead **90**, it is preferred that the stationary blade teeth **96** and the moving blade teeth **100** are positioned so that the respective tips of the teeth of each blade are in close alignment with each other.

[0070] As is the case with the nose trimmer workhead **16** and the dry shaver workhead **60**, when the user desires to exchange the hair trimmer workhead **90** for another, the thumbscrew **54** is loosened through manipulation of the knob **56**. The workhead **90** is then disengaged from the trimmer housing **12** and the drive shaft **18** and the eccentric cam **20** through axial displacing movement.

[0071] Referring now to FIGS. **13-24**, an alternate embodiment for the support brackets **42**, **78**, **102** described above is depicted. Instead of using the thumbscrew **54** and associated components, the alternate support bracket is generally designated **140**, and includes two portions, a base portion **142** fixed to the cutting end **14** of the trimmer housing, and a releasable spring clip portion **144** that includes the first end of the support bracket and is disengageable from the base portion by user manipulation in a twisting motion without the use of tools. Construction of the respective cutting workheads **16**, **60** and **90** is the same, and components of those workheads using the support bracket **140** are designated with reference numbers corresponding to those discussed previously.

[0072] Referring now to FIGS. **13-16**, more specifically the nose trimmer workhead **16** described previously in relation to FIGS. **1-4** is shown with the alternate support bracket **140**. Included on the base portion **142** of the bracket **140** is a second end **146** which is secured within the clevis recess **80** of the trimmer housing **12** using a fastener **148** such as a screw or the like. Also included on the support bracket base portion **142** is an angled mid portion **58**.

[0073] Referring now to FIG. **14**, an upper end **150** of the base portion **142** is generally “Y”-shaped, having a pair of laterally spaced arms **152** defining a recess **154**. A locating opening **156**, preferably polygon-shaped, is positioned between the recess **154** and the angled mid-portion **58**, in a portion of the upper end **150** forming a vertical base of the “Y”-shape.

[0074] Turning now to the spring clip portion **144**, a spring connector **158** includes a lug **160** configured for matingly engaging the locating opening **156**, a depending formation **162** configured for matingly engaging the recess **154**, and at least one, and preferably a pair of mounting openings **164**. Fasteners **166**, preferably screws or the like, secure a workpiece bracket **168** to a spring **170** and to the mounting openings **164** of the spring clip portion **144**. Included on the spring **170** is at least one and preferably a pair of depending spring tabs **172** which engage and are urged against the arms **152**. Similar to the loop **46**, a workpiece ring **174** on the workpiece bracket **168** at least partially surrounds the stationary blade **38** of the nose trimmer workhead **16**.

[0075] Thus, the nose trimmer workhead **16**, the workpiece bracket **168**, the spring **170** and the spring connector **158** are releasably engageable as a unit on the base portion **142** of the support bracket **140**. Secure operational connection is achieved through engagement of the spring tabs **172** on the arms **152**, the depending formation **162** in the recess **154**, and the lug **160** in the locating opening **156**. In addition, the workhead **16** is operationally engaged on the eccentric cam **20** at the formation **22** as described previously. The user is able to readily attach the nose trimmer workhead **16** with the above-listed components of the support bracket **140** to the trimmer housing **12** using a twisting motion and without the use of tools.

[0076] Referring now to FIGS. **17-20**, the dry shaver workhead **60** described previously in relation to FIGS. **5-8** is shown equipped with the alternate support bracket **140**. Components which are identical to those described above are designated with the same reference numbers. The second end

146 of the support bracket **140** is fastened within the clevis recess **80** as described above.

[0077] A first end or cover **176** closes the workpiece housing **62** and is held in place by the fasteners **84**, which extend through the workpiece housing and secure the spring **170** to the spring connector **158**. Thus, the dry shaver workhead **60**, the first end **176**, the spring **170** and the spring connector **158** are operationally engaged on the base portion **142** as described previously, and are readily disengageable from the base portion **142** using a twisting motion by the user without the use of tools.

[0078] Referring now to FIGS. **21-24**, the hair trimmer workhead **90**, described above in relation to FIGS. **9-12** is shown with the alternate support bracket **140**. The fasteners **110** secure the stationary blade **94**, the blade guide **112**, the spring **170** and the spring connector **158** to each other as a unit. Thus, the hair trimmer workhead **90**, including the bladeset **92** and the spring clip portion **144** are operationally engaged on the base portion **142** as described previously, and are readily disengageable from the base portion **142** using a twisting motion by the user without the use of tools.

[0079] Referring now to FIGS. **25-27**, the sequential operation of user-manipulated disengagement of the hair trimmer workhead **90** from the alternate spring grip support bracket **140** is shown. In FIG. **25**, the workhead **90** is shown in an operational position. In FIG. **26**, the user has exerted downward pressure on the workhead **90** on the end opposite the blades **94**, **98**, which disengages the spring force exerted by the spring tabs **172**, disengages the locating lug **160** from the locating opening **156**, the depending formation **162** from the recess **154** and the cam follower **120** from the eccentric cam **20**. In FIG. **27**, in view of the operational disengagement of the workhead **90** from the trimmer housing **12**, the workhead is twisted out of position from the trimmer for storage.

[0080] Regardless of which of the support brackets **42**, **78**, **102** or **140** are used, it is contemplated that a single trimmer housing **12** is provided as an assembly with multiple cutter workheads, including the nose trimmer workhead **16**, the dry shaver workhead **60** and the hair trimmer workhead **90**. Other trimmer workheads are contemplated as are known in the art. Thus, the user is provided with a choice of workheads **16**, **60**, **90** which can be readily exchanged while using only a single trimmer housing **12**.

[0081] Referring now to FIGS. **28-35**, another feature of the present hair trimmer **10** is that it is waterproof for use while showering or bathing or rinseable for cleaning. In the preferred embodiment, this waterproofing is achieved by encasing inner workings of the trimmer within a tubular sleeve of shrink wrap film. Heat is applied to the film, which then shrinks and forms around the trimmer components, sealing them from water damage.

[0082] More specifically, a chassis, generally designated **180** is constructed and arranged for placement within the trimmer housing **12** and includes a motor end **182** corresponding to the drive end **14**, and an opposite charge end **184**. Preferably made of injection molded plastic or similar material as known in the art, the chassis **180** includes a plurality of integrally formed mounting formations **186** for retaining in place at least one of the motor **30**, a rechargeable battery **190** and a circuit board **192** having an ON-OFF switch **194**. A radially enlarged disk-like flange, also referred to as a radial flange **196** is affixed to each of the ends **182**, **184**. The drive shaft **18** projects past the flange **196** at the motor end **182** (FIG. **31**), and at least one battery charge pin **198** extends past the flange at the battery end **184**. In one embodiment, the charge pins **198** are insert molded with the flange **196** to provide a rear moisture seal. An optional protective ring **200** encircles the drive shaft **18**, and a similar optional protective cowl **202** surrounds the charge pins **198** (FIG. **31**).

[0083] Referring now to FIGS. **28** and **29**, to achieve the desired waterproofing of the trimmer chassis **180** and associated components, a tubular sleeve of shrink wrap film **204** is inserted so as to surround the chassis. The sleeve **204** is dimensioned to have a diameter just larger than a diameter of the flanges **196**. Also, the sleeve **204** has a length that extends axially past and overlaps both of the flanges **196**. After suitable placement, heat is applied to the sleeve **204** as is known in the art. Conventionally, temperatures in the range of 250-350° F. are applied to cause shrinkage of the

shrink wrap film, which is preferably made of plastic polymer. The heat is typically applied using a heat gun or a heat tunnel and a conveyor. As the sleeve **204** shrinks, it attaches to and seals around the flanges **196**, as well as other components mounted to the chassis **180**. FIG. **29** depicts the chassis **180** with the shrunken sleeve **204** encapsulating the chassis and components, forming an internal waterproof enclosure.

[0084] To protect the ON-OFF switch **194** during the shrink wrap process, it is preferred that an annular protector **206** preferably formed by a segment of flexible tubing be placed around the switch before the sleeve **204** is placed about the chassis. The protector **206** is preferably at least as tall as the switch. Then, as heat is applied, the sleeve **204** shrinks around the tubing **206** to form a flat or even profile across the switch **194**.

[0085] Referring now to FIGS. **31**, **34** and **35**, the motor **30** is secured to the chassis **180** by a designated motor clamp **208** that is preferably integrally formed with the chassis. The clamp **208** is constructed and arranged for exerting a biasing force that urges the motor **30** towards the associated flange **196**. Also, the motor **30** is covered at the drive shaft end by a resilient boot **210** for further protection. Water is prevented from entering the motor **30** at the driveshaft **18** through the use of a resilient, rubber-like shaft seal **212**. Referring now to FIGS. **32** and **33**, the charge pins **198** have one of a linear (FIG. **32**) and an angled (FIG. **33**) configuration.

[0086] Referring now to FIG. **36**, an alternate embodiment of the shrink wrap waterproof chassis **180** described above is generally designated **220**. Components of the embodiment **220** which correspond to the embodiment **180** are designated with identical reference numbers. In addition to the flanges **196**, an additional flange **222** is placed behind the motor **30**. The sleeve **204**, optionally made of heavier gauge plastic than conventional shrink wrap film, is ultrasonically welded to outer edges **224** of the flanges **196** and optionally the flange **222** around the periphery to create the waterproof seal. The sleeve **204** thus loosely engages the chassis **220** at points exclusive of the flanges **196**, **222**. An external rubber or polyethylene seal **226** is secured to an exterior of the motor flange **196** at the motor end **182**, and a relatively thicker seal **228** of similar material is secured at the charge end **184** for sealing the charge pins **198**.

[0087] Referring now to FIGS. **37** and **38**, an alternate embodiment of the hair trimmer **10**, and particularly to the apparatus for securing the various nose trimmer, shaver and hair trimmer workheads **16**, **60** and **90** of FIGS. **1-11** to the trimmer housing **12** is generally designated **230**. Components of the trimmer **230** shared with previous embodiments are designated with identical reference numbers. During implementation of the hair trimmer **10**, it was observed that upon installation of the various workheads **16**, **60**, **90**, upon the user turning the thumbscrew **54** using the knurled knob **56**, it was not always evident when the thumbscrew was fully tightened against the bracket end **48**.

[0088] Accordingly, there was a perceived need for an improved apparatus for securing the workheads **16**, **60**, **90** to the trimmer housing **12**. In a preferred embodiment, the apparatus includes a non-metal, preferably compressible member or collar **232** that is dimensioned for tightly surrounding a reduced diameter base **234** of the knurled knob **56**. In addition, an exterior **236** of the bushing **232** is preferably co-dimensional with a knurled surface **238** of the knurled knob **56**. An axial thickness "T" of the bushing **232** is at least as thick as the base **234**.

[0089] As the user installs the workheads **16**, **60** or **90** and attaches the thumbscrew **54** with the bushing **232**, the composition of the bushing is such that it compresses upon tightening of the thumbscrew **54** into the insert **53** and against the workhead bracket end **48** in the trimmer housing **12**. The compression of the bushing **232** provides a more positive sensation for the user that the thumbscrew **54** has fully tightened against the bracket end **48**.

[0090] A suitable compressible material for the bushing **232** is plastic, and a preferred plastic is Delrin® acetal resin or equivalent. Other materials are contemplated as having suitable dimensional stability, strength and compressibility. In the present trimmer **230**, the relatively low co-efficient of friction of the Delrin® acetal resin member **232** makes it easier to tighten the

thumbscrew **54** by applying more torque to create more clamping (axial load) to secure the blade bracket down.

[0091] A preferred thickness “T” is 0.020 inch greater than a thickness of the base **234** of the knob **56**. It is contemplated that the dimensions of the bushing **232** may vary to suit the application. By making the bushing **232** thicker than the base, the adhesion of the knob **56** due to compression forces is enhanced.

[0092] To enhance friction forces holding the bushing **232** onto the base **234** of the knob **56**, an inner surface **240** of the bushing is optionally provided with at least one radially projecting rib **242**. The number and dimensions of the rib **242** are contemplated as varying to suit the application.

[0093] Referring now to FIGS. **39-41**, an alternate view of the present trimmer motor chassis **180**, **220**, also referred to as a motor pod when assembled and encased in the sleeve of shrink wrap film **204**, is generally designated **250**. Components shared with the embodiments **10**, **60**, **90**, **180** and **220** are designated with identical reference numbers.

[0094] In the embodiment **250**, a main difference from the chassis **180**, **220** is that the flanges **196** have a reduced diameter. As such, the assembled chassis **250** has a smaller profile, which is favored for use in relatively slimmer trimmer housings **12**. However, a drawback of the smaller diameter flanges **196** is that there is a reduced bonding area for the flexible shrink wrap sleeve **204**. Accordingly, there is a concern that the water resistant seal of the chassis **250** may be compromised.

[0095] To enhance the water resistance of this slimmer profile motor pod **250**, the pod is equipped with resilient endcaps **252**. The endcaps **252** are configured with an end wall **254** and an annular skirt **256** projecting from a peripheral edge of the end wall to extend axially along the motor pod/chassis **250** to sufficiently envelop the motor pod such that a positive seal is achieved with the shrink wrap sleeve **204**. While other dimensions are contemplated, each endcap skirt **256** extends axially approximately one inch (2.5 cm) from the end wall **254**.

[0096] An endcap **252** is placed at each end of the chassis **250** adjacent the flanges **196**. The endcaps **252** are not identical, because adjacent the motor **30**, and more particularly at the motor shaft seal **212**, a motor endcap **260**, otherwise identical to the endcap **252**, is provided with an opening **262** (FIG. **41**) dimensioned for accommodating the protective drive ring **200** and the motor shaft seal **212**. The endcap **252** adjacent the charge pins **198** is also provided with an opening (not shown) dimensioned for accommodating the protective charge cowl **202**. Opposite the motor shaft seal end, the trimmer motor **30** is held in place using the motor clips **208** that are preferably integrally molded into the chassis **250**.

[0097] Upon completion of the assembly of the chassis **250**, the endcaps **252**, **260** are placed at respective ends of the chassis, and the flexible shrink wrap sleeve **204** is slid over the chassis so that ends **264** of the sleeve overlap the skirts **256** to create a seal once heat is applied. In preferred embodiments, the flexible shrink wrap sleeve **204** is made of waterproof-grade material, which has an adhesive on an inner surface **266**. This adhesive has been found to enhance the bonding of the sleeve **204** to the endcap skirts **256**. Upon application of heat, the flexible shrink wrap sleeve **204** is tightly formed around the motor pod **250** (FIG. **40**), and tightly and sealingly binds to the endcaps **252**, **260**. A water resistant, if not water proof seal around the chassis **250** is thus achieved.

[0098] The assembled chassis **250** with the endcaps **252**, **260** and the shrink wrap sleeve **204** applied is also referred to as an encased drive module, since it contains the significant trimmer operational components, namely the motor **30**, the battery **190**, the operational circuit board **192**, and provides them in a single, water-tight assembly.

[0099] Another advantage of the endcaps **252**, **260** is the resilient, rubber-like material from which they are constructed. In a preferred embodiment, the endcaps **252**, **260** are made of silicone rubber, however other resilient materials are contemplated. The endcap material is compressible, and upon assembly of the motor pod **250** into the trimmer housing **12**, the endcaps **252**, **260** enhance operational noise reduction.

[0100] While a particular embodiment of the present outliner hair trimmer with interchangeable workheads has been described herein, it will be appreciated by those skilled in the art that changes and modifications may be made thereto without departing from the invention in its broader aspects and as set forth in the following claims.

Claims

1. A hair trimmer, comprising: a trimmer housing having a drive end; a drive shaft projecting from said drive end; a chassis having one end associated with said drive shaft and an opposite end having at least one projecting charging pin; a radial flange at each of said ends; and a tubular sleeve of heat deformable plastic film dimensioned for slidably surrounding said chassis and extending axially at least to said flanges, such that upon application of heat, said sleeve deforms around said chassis and forms a watertight enclosure.
 2. The hair trimmer of claim 1, wherein said flanges form sealing attachment points for said tubular sleeve, wherein said sleeve is attached to said chassis by one of heat shrinking and welding to said flanges.
 3. The hair trimmer of claim 1, further including a switch secured to said housing, and an annular protector surrounding said switch and forming a guide for, and provide a flat profile for said switch as said sleeve is heated and sealingly engages portions of said trimmer.
 4. The hair trimmer of claim 1, further including a pair of endcaps, each placed, prior to installation of said tubular sleeve, at a corresponding end of said chassis and including a skirt projecting axially along said chassis to form a bonding point for said tubular sleeve.
 5. The hair trimmer of claim 4, wherein at said drive shaft end, said endcap is provided with an opening accommodating said drive shaft and a drive shaft seal.
 6. The hair trimmer of claim 4, wherein at each said endcap, said skirt extends axially at least one inch.
 7. The hair trimmer of claim 4 wherein said endcaps are made of resilient material.
 8. A hair trimmer, comprising: a trimmer housing having a drive end; a drive shaft projecting from said drive end; a chassis having one end associated with said drive shaft and an opposite end having at least one projecting charging pin; a radial flange at each of said ends; an endcap at each of said ends, each said endcap having an end wall from which axially extends an annular skirt, said annular skirt dimensioned to enclose said corresponding chassis end; and a tubular sleeve of heat deformable plastic film dimensioned for slidably surrounding said chassis and extending axially at least to said flanges, such that upon application of heat, said sleeve deforms around said chassis and said endcaps and forms a watertight enclosure.
 9. An encased drive module, comprising: a chassis having one end associated with a drive shaft and an opposite end having at least one projecting charging pin; a radial flange at each of said ends; an endcap of resilient material enclosing each said end of said chassis; a tubular sleeve of heat deformable plastic shrink wrap film dimensioned for slidably surrounding said chassis and said endcaps and extending axially at least to said flanges, such that upon application of heat, said sleeve deforms and shrinks around said chassis and said endcaps and forms a watertight enclosure.
 10. The encased drive module of claim 9, wherein said chassis includes a plurality of integrally formed mounting formations for retaining components mounted to said chassis, and wherein said tubular sleeve attaches to and seals around said flanges and said components mounted to said chassis.
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